

Supplementary Material

Apalutamide plus androgen deprivation therapy in clinical subgroups of patients with metastatic castration-sensitive prostate cancer: a subgroup analysis of the randomised clinical TITAN study

Axel S. Merseburger, Neeraj Agarwal, Amitabha Bhaumik, Florence Lefresne, Lawrence Karsh, Andrea Juliana P. de Santana Gomes, Álvaro Juárez Soto, Robert Given, Sabine D. Brookman-May, Suneel D. Mundle, Sharon A. McCarthy, Hirotsugu Uemura, Simon Chowdhury, Kim N. Chi, Anders Bjartell

Contents

Supplementary Methods

TITAN was a multicenter, randomized, placebo-controlled phase 3 study that enrolled 1052 patients at primary, secondary and tertiary health care centers and community centers. Patients were diagnosed with castration-sensitive prostate cancer (CSPC) with distant metastatic disease, including at least one lesion on bone scan, and an Eastern Cooperative Oncology Group performance status of 0 or 1. Eligible patients were required to have documented adenocarcinoma of the prostate and distant metastatic disease per conventional imaging with or without visceral or lymph-node involvement. TITAN patients were required to be castration sensitive (i.e., patients were not receiving androgen-deprivation therapy [ADT] at the time of disease progression). Patients were stratified according to Gleason score at diagnosis (≤7 vs >7), geographic region (North America and European Union versus all other countries) and previous treatment with docetaxel (yes vs no) and then randomised 1:1 to receive oral apalutamide 240 mg/day (n = 525) or placebo (n = 527) added to concurrent ADT. The interactive web response system (IWRS) randomly balanced stratification criteria and generated a unique treatment code that dictated patient's treatment, using randomly permuted blocks. The patients, staff and sponsor representatives were unaware of treatment allocations. IWRS maintained generated patient codes.

Prior treatment for metastatic CSPC (mCSPC) was limited to previous docetaxel (up to six cycles, with the last dose ≤2 months before random assignment and with no evidence of progression during treatment or before random assignment), ADT for ≤6 months and primary radiation or surgical intervention for localised disease completed at least 1 year before random assignment. The study was approved by review boards of participating institutions and was conducted in accordance with International Council for Harmonisation guidelines and according to Declaration of Helsinki principles. All the patients provided written informed consent.

Dual primary end-points of TITAN were radiographic progression-free survival (rPFS) and overall survival (OS). rPFS was defined as time from randomisation to first documented progressive disease by conventional imaging or death. The study assessed conventional imaging based on Response Evaluation Criteria In Solid Tumors (RECIST), version 1.1, modified by Prostate Cancer Working Group 2 (PCWG2) criteria [1]. Overall survival was defined as the time from randomisation to the date of death from any cause.

TITAN secondary end-points were the time to cytotoxic chemotherapy defined as the time from randomisation to initiation of cytotoxic chemotherapy, time to pain progression as assessed by average increase in 2 points from baseline in Brief Pain Inventory–Short Form, time to chronic opioid use defined as the time from randomisation to chronic opioid use and time to skeletal-related event defined as the time from randomisation to first observation of a skeletal-related event (symptomatic pathologic fracture, spinal cord compression, radiation to bone or surgery to bone).

rPFS was assessed first. As the difference between the apalutamide and placebo groups was statistically significant, the alpha was recycled to OS on the basis of the fallback method with an overall type I error of 5%. The study was considered a success if at least one of the two primary end points is statistically significant. TITAN was designed with ≥85% power to detect a 33% reduction in the hazard of radiographic progression with a two-tailed significance level of 0.0005 and with ≈80% power to detect a 25% reduction in the hazard of death with a two-tailed significance level of 0.045 for patients receiving apalutamide. The first interim analysis was performed at ≈50% of total required deaths. For the protocol-defined final updated OS analysis, 410 events were required. Evaluation of secondary end-points was performed in the hierarchical order, each with an overall two-sided significance level of 0.05: time to cytotoxic chemotherapy, time to pain progression, time to chronic opioid use and time to skeletal-related event. Serum prostate-specific antigen was assessed at screening, on day 1 of each treatment cycle until cycle 13, every other cycle until cycle 25 and every four cycles until the end-of-treatment visit. Treatment-emergent adverse events (TEAEs) were defined as AEs that occurred on or after first dose of the study drug through one cycle, defined as 30 days after the last study treatment. TEAEs were coded using Medical Dictionary for Regulatory Activities versions 20.0 and graded according to the National Cancer Institute Common Terminology Criteria for Adverse Events version 4.03.

Table S1. Events and patient at risk across subgroups

End-point Treatment + ADT	Events/patients at risk, n/*n* (%)			
	Synchronous /high-volume	Synchronous /low-volume	Metachronous /high-volume	Metachronous /low-volume
OS				
Apalutamide	112/273 (41%)	28/138 (20%)	13/32 (41%)	7/53 (13%)
Placebo	156/294 (53%)	43/147 (29%)	15/28 (54%)	14/31 (45%)
rPFS				
Apalutamide	89/273 (33%)	19/138 (14%)	12/32 (38%)	5/53 (9.4%)
Placebo	151/294 (51%)	45/147 (31%)	15/28 (54%)	8/31(26%)
PFS2				
Apalutamide	114/273 (42%)	29/138 (21%)	13/32 (41%)	7/53 (13%)
Placebo	164/294 (56%)	46/147 (31%)	15/28 (54%)	14/31 (45%)
Time to PSA progression				
Apalutamide	97/273 (36%)	16/138 (12%)	9/32 (28%)	8/53 (15%)
Placebo	213/294 (72%)	87/147 (59%)	17/28 (61%)	15/31 (48%)
Time to castration resistance				
Apalutamide	132/273 (48%)	31/138 (22%)	10/32 (31%)	9/53 (17%)
Placebo	230/294 (78%)	92/147 (63%)	20/28 (71%)	18/31 (58%)

End-point Treatment group	Events/patients at risk, n/*n* (%)			
	Metachronous	Synchronous	High-volume	Low-volume
OS				
Apalutamide	20/85 (24%)	140/411 (34%)	134/325 (41%)	36/200 (18%)
Placebo	29/59 (49%)	199/441 (45%)	175/335 (52%)	60/192 (31%)
rPFS				
Apalutamide	17/85 (20%)	108/411 (26%)	109/325 (34%)	25/200 (13%)
Placebo	23/59 (39%)	196/441 (44%)	173/335 (52%)	58/192 (30%)
PFS2				
Apalutamide	20/85 (24%)	143/411 (35%)	119/325 (37%)	34/200 (17%)

Placebo	29/59 (49%)	210/441 (48%)	152/335 (45%)	48/192 (25%)
Time to PSA progression				
Apalutamide	17/85 (20%)	113/411 (28%)	114/325 (35%)	24/200 (12%)
Placebo	32/59 (54%)	300/441 (68%)	234/335 (70%)	110/192 (57%)
Time to castration resistance				
Apalutamide	19/85 (22%)	163/411 (40%)	151/325 (47%)	40/200 (20%)
Placebo	38/59 (64%)	322/441 (73%)	257/335 (77%)	118/192 (62%)

ADT: androgen-deprivation therapy; OS: overall survival; PFS2: second progression-free survival; PSA: prostate-specific antigen; rPFS: radiographic progression-free survival.

Fig. S1. CONSORT diagram.

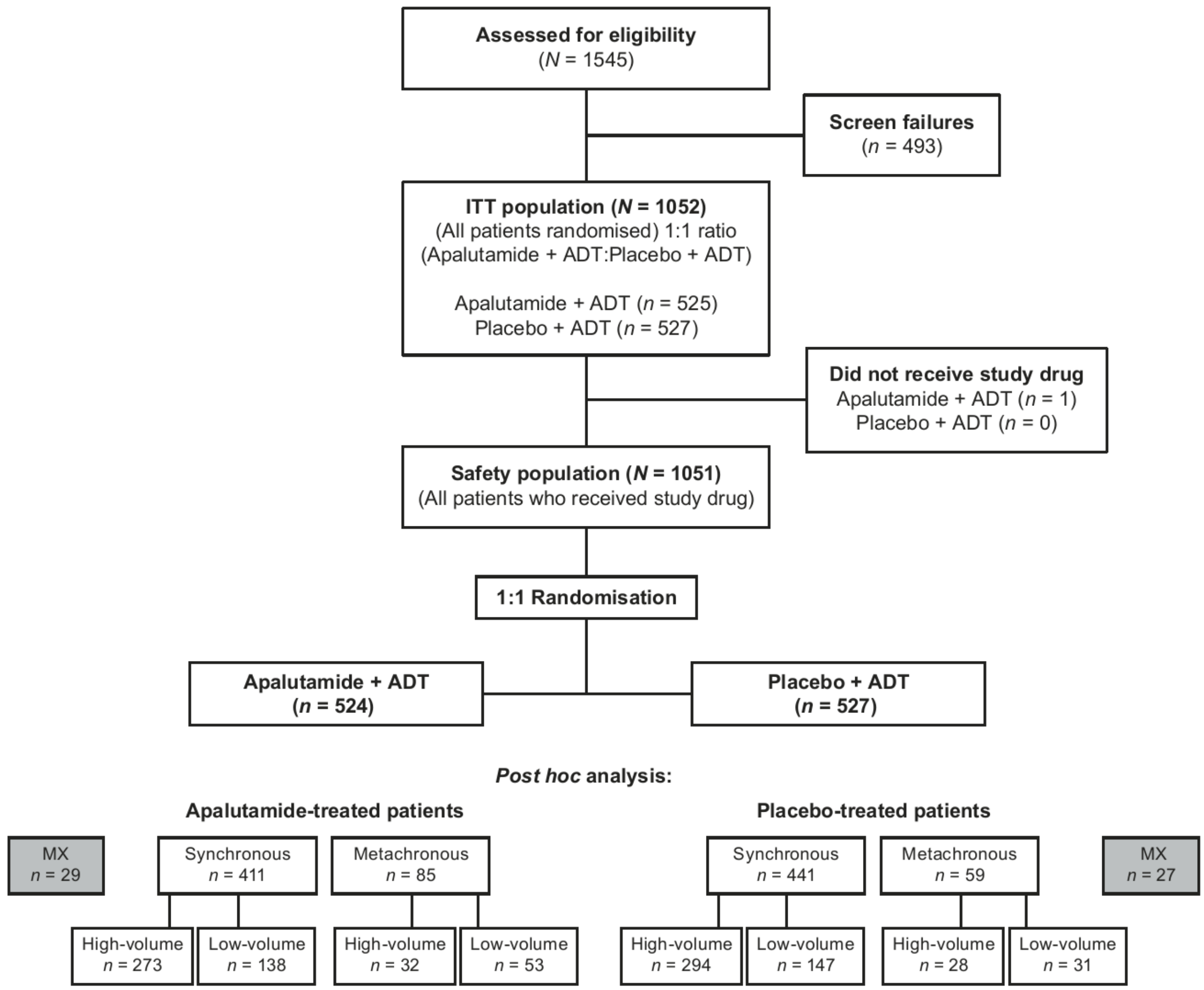

Fig. S2. Treatment effect of apalutamide on second progression-free survival (PFS2). Synchronous/high-volume (A), synchronous/low-volume (B), metachronous/high-volume (C), and metachronous/low-volume (D) disease. The curves are truncated for instances in which the number at risk in a group was <5.

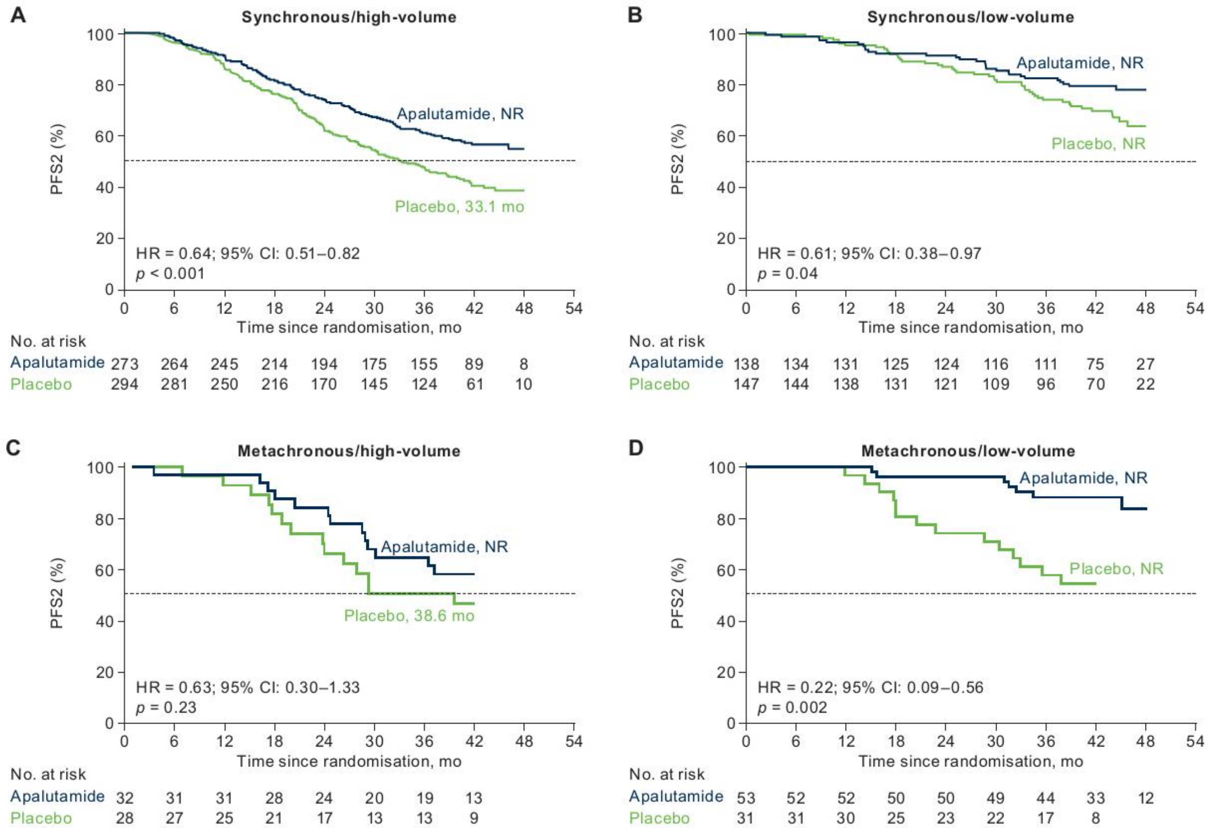

Fig. S3. Treatment effect of apalutamide on time to castration resistance.
Synchronous/high-volume (A), synchronous/low-volume (B), metachronous/high-volume (C), and metachronous/low-volume (D) disease. The curves are truncated for instances in which the number at risk in a group was <5.

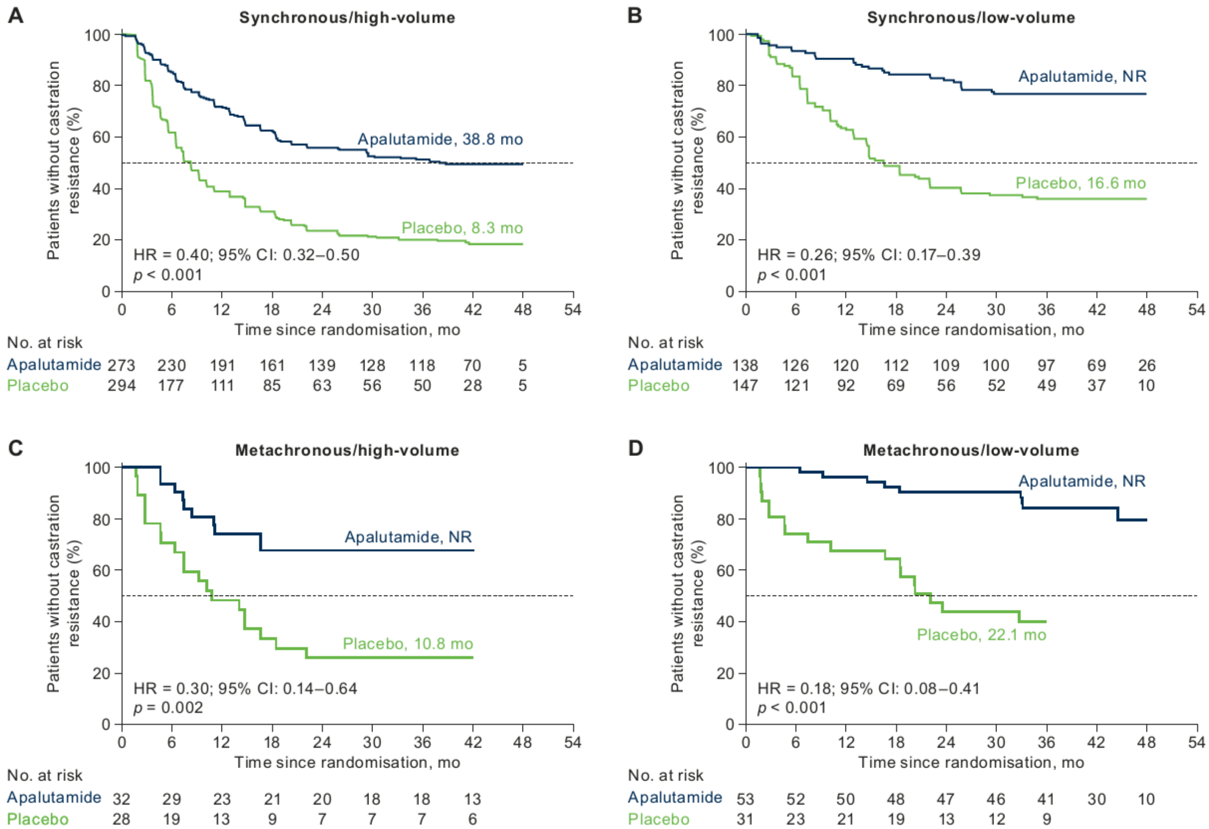

Fig. S4. Treatment effect of apalutamide on time to PSA progression.
Synchronous/high-volume (A), synchronous/low-volume (B), metachronous/high-volume (C), and metachronous/low-volume (D), synchronous (E), metachronous (F), high-volume (G), and low-volume (H) disease subgroups are shown. The curves are truncated for instances in which the number at risk in a group was <5.

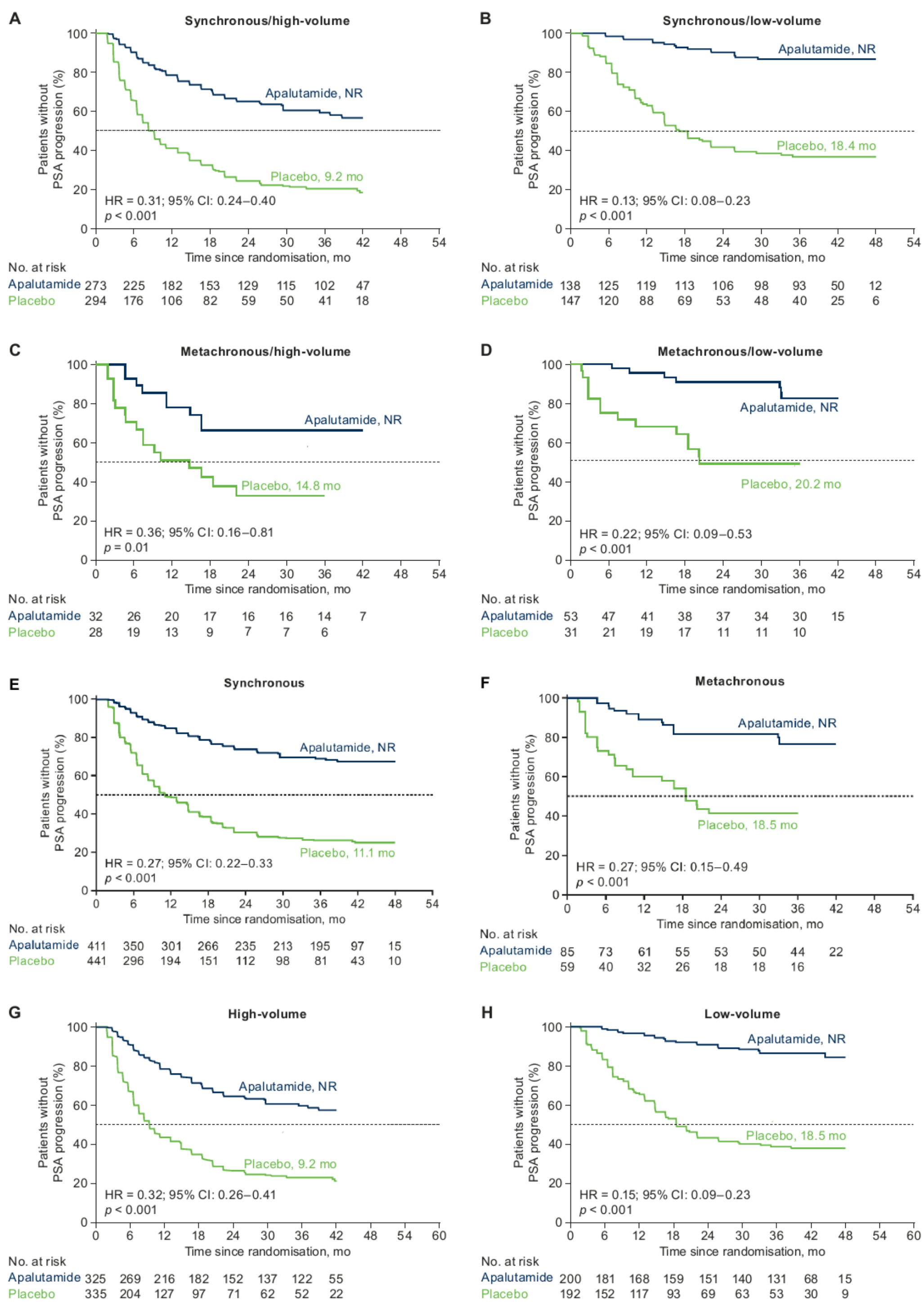

Fig. S5. Treatment effect of apalutamide on confirmed best PSA decline at any time during the study.
Synchronous/high-volume (A), synchronous/low-volume (B), metachronous/high-volume (C), and metachronous/low-volume (D) disease. PSA decline of ≥50% or ≥90% from baseline or as PSA ≤0.2 ng/mL is shown.

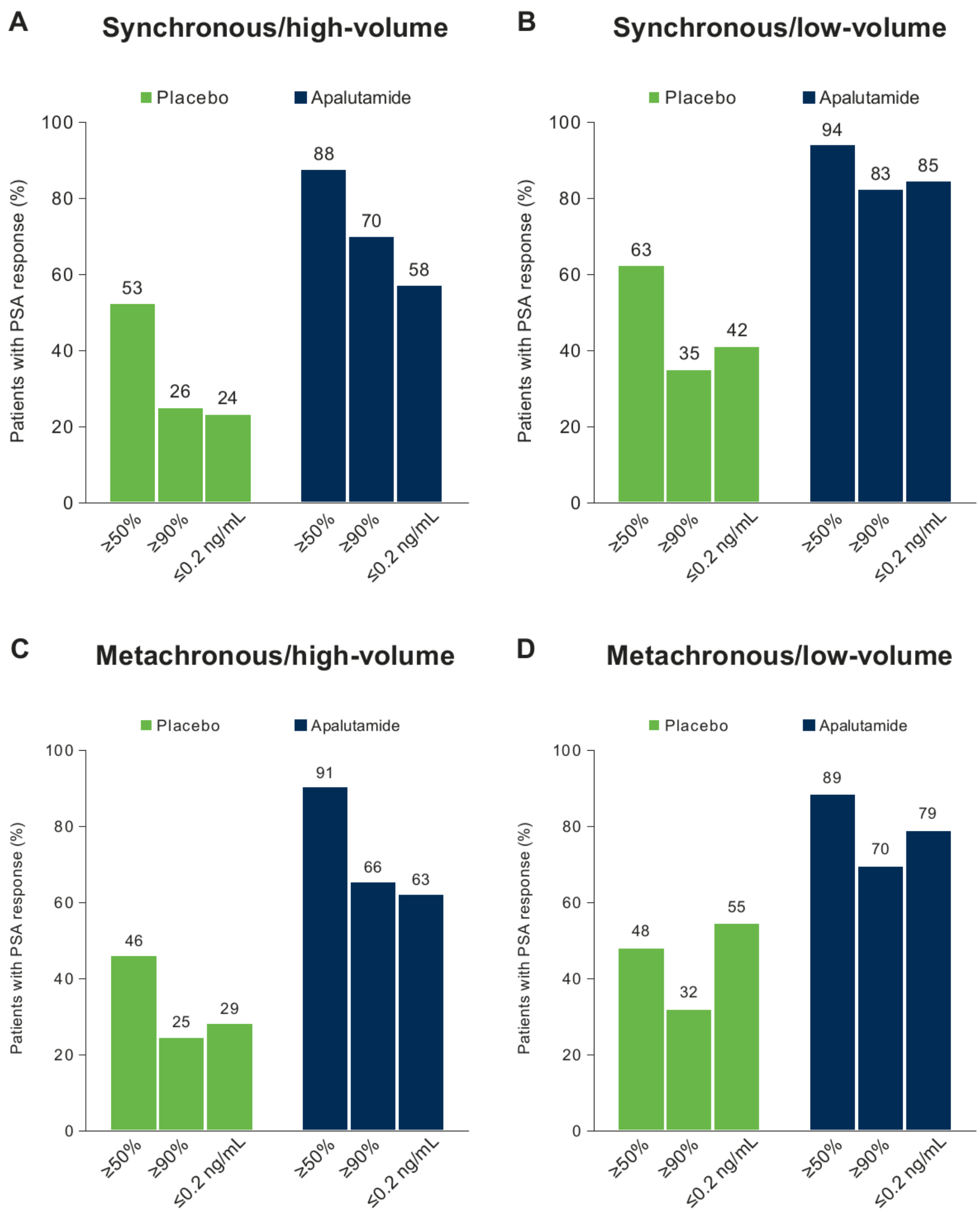

Fig. S6. Treatment effect of apalutamide on OS, rPFS, PFS2 and time to castration resistance in patients with synchronous or metachronous disease.
OS (A, B), rPFS (C, D), PFS2 (E, F) and time to castration resistance (G, H) were analysed in synchronous (A, C, E and G) and metachronous (B, D, F and H) subgroups. The curves are truncated for instances in which the number at risk in a group was <5.

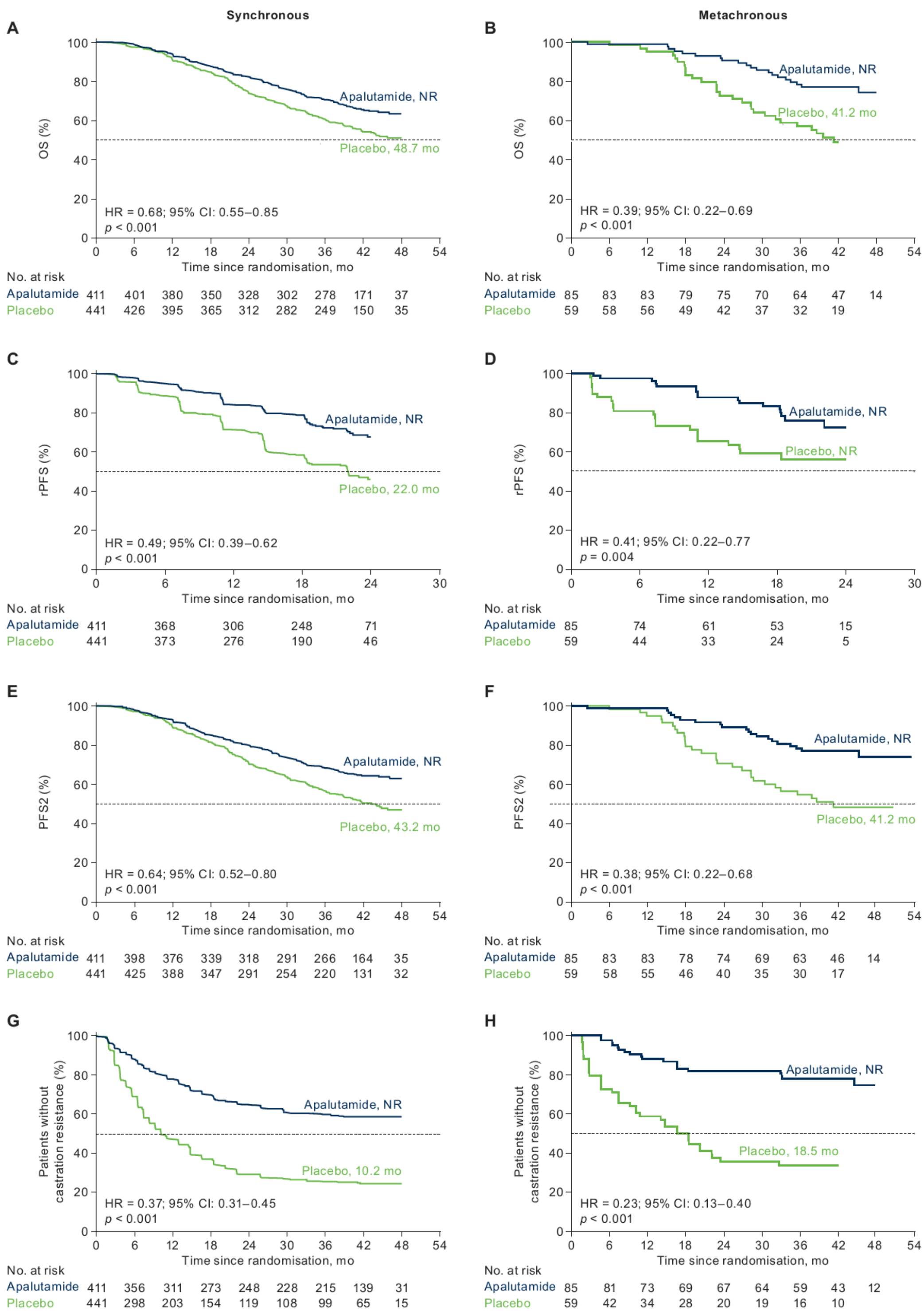

Fig. S7. Treatment effect of apalutamide on OS, rPFS, PFS2 and time to castration resistance in patients with high- or low-volume disease.
OS (A, B), rPFS (C, D), PFS2 (E, F) and time to castration resistance (G, H) were analysed in high-volume (A, C, E and G) and low-volume (B, D, F and H) subgroups. The curves are truncated for instances in which the number at risk in a group was <5.

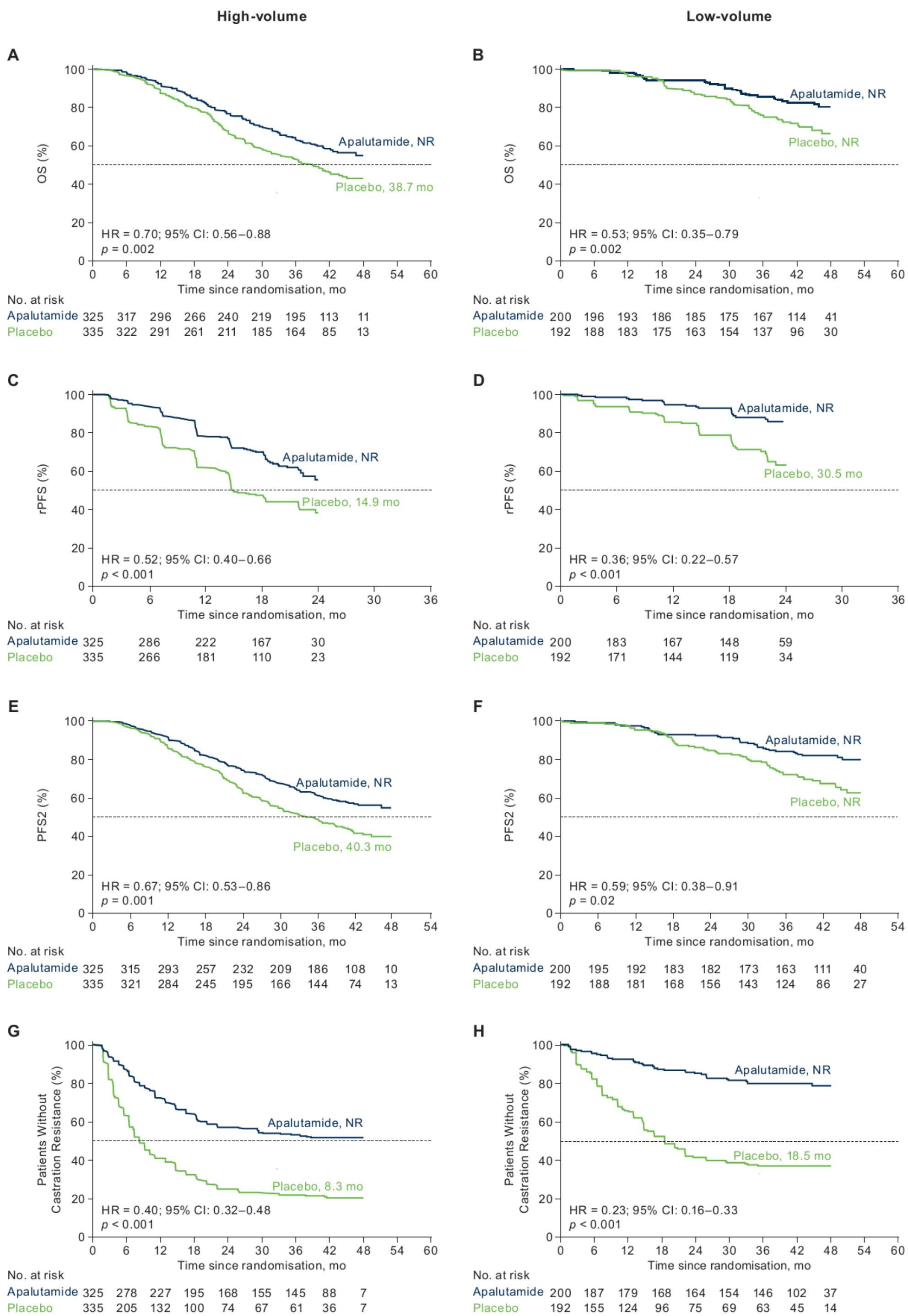

Fig. S8. The effect of oligo- and polymetastatic disease on clinical with ADT alone. Overall survival (OS) (A), radiographic progression-free survival (rPFS) (B), second progression-free survival (PFS2) (C) and time to castration resistance (D). The curves are truncated for instances in which the number at risk in a group was <5. Mets: metastases; NR: not reached.